

An Improved Differential Evolution-Based Hand-Eye Calibration Optimization Algorithm with Elastic Net Regularization: Supplementary File

I. ADDITIONAL TABLES

TABLE S.I
JOINT ANGLES OF THE INDUSTRIAL ROBOT MOVEMENT

No.	q_1	q_2	q_3	q_4	q_5	q_6	No.	q_1	q_2	q_3	q_4	q_5	q_6
1	-10.52	-8.05	-6.52	-1.60	89.55	-4.10	21	-11.40	-11.88	-6.52	-5.72	92.79	-4.10
2	-16.48	-8.05	-6.52	-1.60	89.55	-4.10	22	-11.40	-11.88	-6.52	-5.72	88.61	-4.10
3	-13.21	-7.52	-6.52	-1.60	89.55	-4.10	23	-11.40	-11.88	-6.52	-7.77	88.61	-4.10
4	-8.24	-7.52	-6.52	-1.60	89.55	-4.10	24	-11.40	-11.88	-6.52	2.35	88.61	-4.10
5	-5.32	-7.52	-6.52	-1.60	89.55	-4.10	25	-11.40	-11.88	-6.52	2.35	87.42	-4.10
6	-4.20	-7.62	-6.52	-1.60	89.55	-4.10	26	-11.40	-11.88	-6.52	2.35	92.60	-4.10
7	-3.88	-9.90	-6.52	-1.60	89.55	-4.10	27	-11.40	-11.88	-6.52	4.17	92.60	-4.10
8	-10.18	-9.90	-6.52	-1.60	89.55	-4.10	28	-11.40	-11.88	-6.52	4.17	93.63	-4.10
9	-13.68	-9.90	-6.52	-1.60	89.55	-4.10	29	-11.40	-11.88	-6.52	4.17	89.49	-4.10
10	-16.51	-11.88	-6.52	-1.60	89.55	-4.10	30	-11.40	-11.88	-6.52	-1.48	89.49	-4.10
11	-11.40	-11.88	-6.52	-1.60	89.55	-4.10	31	-11.40	-11.88	-6.52	-1.48	93.91	-4.10
12	-11.40	-11.88	-6.52	-1.60	93.12	-4.10	32	-11.40	-11.88	-6.52	-6.45	93.79	-4.10
13	-11.40	-11.88	-6.52	2.79	93.12	-4.10	33	-11.40	-11.88	-6.52	-6.45	88.66	-4.10
14	-11.40	-11.88	-6.52	-4.13	93.12	-4.10	34	-11.40	-11.88	-6.52	2.34	88.66	-4.10
15	-11.40	-11.88	-6.52	-4.13	88.56	-4.10	35	-11.40	-11.88	-6.52	2.34	94.15	-4.10
16	-11.40	-11.88	-6.52	-6.95	88.56	-4.10	36	-11.40	-11.88	-6.52	4.69	93.65	-4.10
17	-11.40	-11.88	-6.52	1.67	88.56	-4.10	37	-16.25	-11.43	-6.52	4.69	93.65	-4.10
18	-11.40	-11.88	-6.52	1.67	92.98	-4.10	38	-18.82	-11.43	-6.52	4.69	93.65	-4.10
19	-11.40	-11.88	-6.52	4.23	92.98	-4.10	39	-22.21	-11.45	-6.52	4.69	93.65	-4.10
20	-11.40	-11.88	-6.52	-0.19	92.98	-4.10	40	-24.75	-11.58	-6.52	4.69	93.65	-4.10

TABLE S.II
ALGORITHM PARAMETER SETTINGS

Algorithm	Parameter Settings
IDER	$N_P = 100$; $T = 120$; Scaling factor $F = 0.6$; Crossover rate $CR = 0.9$.
PSO	$N_P = 100$; $T = 120$; Inertia weight $\omega = 0.6$; Learning factors $c_1 = 1.2$, $c_2 = 1.2$.
DE	$N_P = 100$; $T = 120$; Scaling factor $F = 0.6$; Crossover rate $CR = 0.9$.
GA	$N_P = 100$; $T = 120$; Crossover rate: 0.9; Mutation rate: 0.08; Tournament selection size: 3.
QPSO	$N_P = 100$; $T = 120$; Contraction-expansion coefficient: 1; Mean optimal coefficient: 0.06.
CS	$N_P = 100$; $T = 120$; Abandonment probability $pa = 0.15$; Step size scaling factor: 0.6; Characteristic exponent $\beta = 1.8$.
GWO	$N_P = 100$; $T = 120$; Initial convergence factor: 1.8; Final convergence factor: 0.
SCA	$N_P = 100$; $T = 120$; Initial attenuation factor: 5; Final attenuation factor: 0.
ABC	$N_P = 100$; $T = 120$; Limit: 15; Scout bee trigger frequency: 12.
FA	$N_P = 100$; $T = 120$; Maximum attractiveness: 1; Light absorption coefficient: 0.8; Random step factor: 0.7.

This is the supplementary material for the paper. It contains extra tables and figures concerning the model structure and experimental results.

Algorithm 1 IDER Algorithm.**Input:** x_0, A, B

INITIALIZATION:

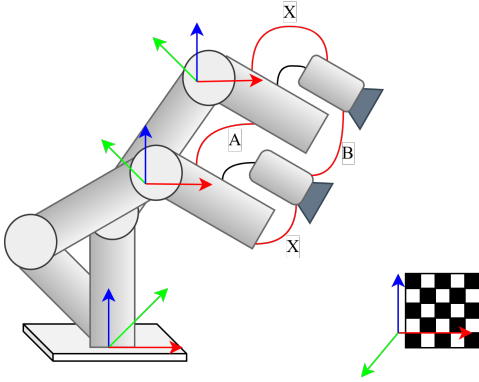
1. **Initialize:** F, CR, T, N_P
2. **Initialize:** $x_{de} = x_0, f(X_{de}) = f(X_0)$

OPTIMIZATION STEP:

3. **for** $t = 1$ **to** T
4. Update x_i with Eq. (23)
5. Calculate v_i with Eq. (26)
6. Update u_i with Eq. (28)
7. Update u_i with Eq. (30)
8. Calculate f_{ui} with Eq. (25)
9. Update $x_{(t+1)}$ with Eq. (31)
10. **end for**
11. $x_{de} = x_{(t+1)}$
12. **for** $t = 1$ **to** T
13. **if** $|E_1| \geq 1$
14. Update $P_i^{(t+1)}$ with Eq. (38)
15. **else**
16. Update $P_i^{(t+1)}$ with Eq. (39)
17. **end if**
18. Calculate $F(\alpha_{\text{opt}}, \lambda_{\text{opt}})$ with Eq. (34)
19. Update (α^*, λ^*) with Eq. (42)
20. Calculate x_{en} with Eq. (33)
21. Update X_{en}
22. Calculate RMSE with Eq. (43)
23. **end for**

Output: X_{en}, RMSE

II. ADDITIONAL FIGURES

Fig. S.1. Formulating the $AX=XB$ problem.